

Sources of Resistance to Muscle Contraction

The most common sources of resistance for strength training exercises are gravity, **inertia**, friction, fluid resistance, and **elasticity**. This section provides information on how we overcome these forms of resistance during acts of daily living, exercise, and sport. An understanding of the principles behind exercise devices using the various forms of resistance can provide insight into their effectiveness and applicability.

What Goes Up Must Come Down: Gravity and Vertical Jump Performance

In sport, athletes launch themselves into the air. Jump height depends on takeoff vertical velocity (TOV). To jump higher, athletes must leave the ground faster because it takes gravity longer to reduce velocity to 0. Calculate jump height by squaring TOV and dividing it by two times gravity. To jump 0.55 meters requires a TOV of $3.285 \text{ m} \cdot \text{s}^{-1}$. If calculating the TOV needed to jump a certain height, first multiply the new jump height by two times gravity before calculating its square root: $0.55 \times 2 \times \text{gravity} = 3.285 \text{ m} \cdot \text{s}^{-1}$.

Gravity

The downward force on an object from the pull of **gravity**, the interaction of mass and gravity, is equal to the object's mass times the local acceleration due to gravity:

$$F_g = m \times a_g \quad (2.8)$$

where F_g is the force due to gravity (same as the object's weight, measured in newtons [N]), m is the object's mass (measured in kilograms [kg]), and a_g is the local acceleration due to gravity (measured in meters per second per second [$\text{m} \cdot \text{s}^{-2}$]). The acceleration due to gravity can vary by geographic location. Weighing a barbell on a calibrated spring or electronic scale shows

its actual weight. A balance scale determines only the object's mass, so its weight (F_g) must be calculated using equation 2.8 if a spring or electronic scale is not available.

Popular terminology for *weight* and *mass* is often incorrect and used interchangeably with correct terminology. As explained earlier, the correct measurement unit for mass, which describes the load presented to us by, for example, barbells and dumbbells, is the kilogram (kg). Therefore, a barbell loaded to 85 kilograms should be referred to as "the barbell has a mass of 85 kilograms" or "the barbell is 85 kilograms" without referring to weight. Use of *pounds* as a measurement unit of mass is incorrect because it is a measurement unit once used for force. However, we know that the correct measurement unit for both weight and force is the newton and that this describes the interaction of mass and gravity ([table 2.1, page 37](#)). While the weight of the 85-kilogram barbell on planet Earth is 833.85 newtons, on the moon its weight would be 137.7 newtons, making it feel more like 14 kilograms, even though it did not physically change.

Applications to Resistance Training

The gravitational force on an object always acts downward. Since, by definition, the moment arm by which a force produces torque is perpendicular to the line of action of the force, the moment arm of a weight is always horizontal. Thus, torque due to an object's weight is the product of the weight and the length from the weight to the axis of rotation (joint). During an exercise, although the weight does not change, the length between the weight and a given joint axis can change constantly. When the weight is horizontally closer to the joint, it exerts less resistive torque; when it is horizontally farther from a joint, it exerts more resistive torque. For example, in the biceps curl, the length element is greatest when the forearm is horizontal. Thus, in that position the athlete must exert the greatest muscle torque to support the weight. The moment arm decreases as the forearm rotates either upward or downward away from the

horizontal position, decreasing the resistive torque arising from the weight ([figure 2.7, page 31](#)). When the weight is directly above or below the elbow axis of rotation, there is no resistive torque from the weight.

Exercise technique can affect the resistive torque pattern during an exercise and can shift stress among muscle groups. In the back squat, for example, a more forward inclination of the trunk brings the weight horizontally closer to the knees, thus reducing the resistive torque about the knees that the quadriceps must counteract. At the same time, the weight is horizontally farther from the hip, increasing the resistive torque about the hip that the gluteus and hamstring muscles must counteract. This resistive torque pattern is most often present when the barbell is positioned as low as possible on the upper back (often termed a *low bar squat*); the athlete must incline the trunk relatively far forward to keep the center of mass of body plus bar over the feet, thereby avoiding a fall. Because the bar is then horizontally far from the hip and close to the knee, stress is focused on the hip extensors and, to a lesser extent, on the knee extensors. The opposite of this resistive torque pattern would occur in a high bar squat, in which the bar is placed higher up on the back closer to the neck. As a result of this bar positioning, the torque distribution increases the resistive torque about the knees and concomitantly reduces the resistive torque about the hip (in contrast to what occurs with the low bar squat).

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Weight-Stack Machines

As with free weights, gravity is the source of resistance for weight-stack machines. However, by means of pulleys, cams, cables, and gears, these machines provide increased control over the direction and pattern of resistance. Both free weights and stack machines

have advantages and disadvantages. The following are some of the advantages of the stack machine:

- *Safety.* The likelihood of injury as a result of being hit by, tripping over, or being trapped under a weight is reduced. Less skill is required to maintain control of a weight stack than a free weight.
- *Design flexibility.* Machines can be designed to provide resistance to body movements that are difficult to resist with free weights (e.g., lat pulldown, hip adduction and abduction, leg curl). To some extent, the pattern of resistance can be engineered into a machine.
- *Ease of use.* Many people who fear that they lack the coordination or technique to safely lift free weights feel confident when using machines. Also, it is quicker and easier to select a weight by inserting a pin in a stack than by mounting plates on a bar.

Advantages of free weights include the following:

- *Total body training.* Free weight exercises are often performed in the standing position with the weight supported by the entire body, taxing a larger portion of the body's musculature and skeleton than a weight-stack machine would. Such weight-bearing exercise promotes greater bone mineralization, helping to prevent osteoporosis in later life (18). Moreover, the movement of a free weight is constrained by the athlete rather than by a machine, requiring muscles to work in stabilization as well as support. A **structural exercise**, such as the power clean or the snatch, is particularly useful in providing training stimulus for a major portion of the body's musculature.
- *Simulation of real-life activities.* The lifting and acceleration of objects represent a major part of sport and other physically demanding activities. Machines tend to isolate single muscle groups; the lifting of free weights involves the more natural coordination of several muscle groups.

Nautilus Sports/Medical Industries popularized the concept of tailoring resistive torque through the range of joint motion by creating an exercise machine that uses a cam of variable radius; this changes the length of the moment arm through which the weight stack acts ([figure 2.14](#)). The rationale was to provide more resistance at points in the range of motion where the muscles could exert greater torque, and less resistance where the muscles could apply less torque. For the system to work as planned, however, the athlete has to move at a constant, slow angular velocity, which is difficult to do consistently. Also, cam-based machines frequently fail to match normal human torque capability patterns (12).

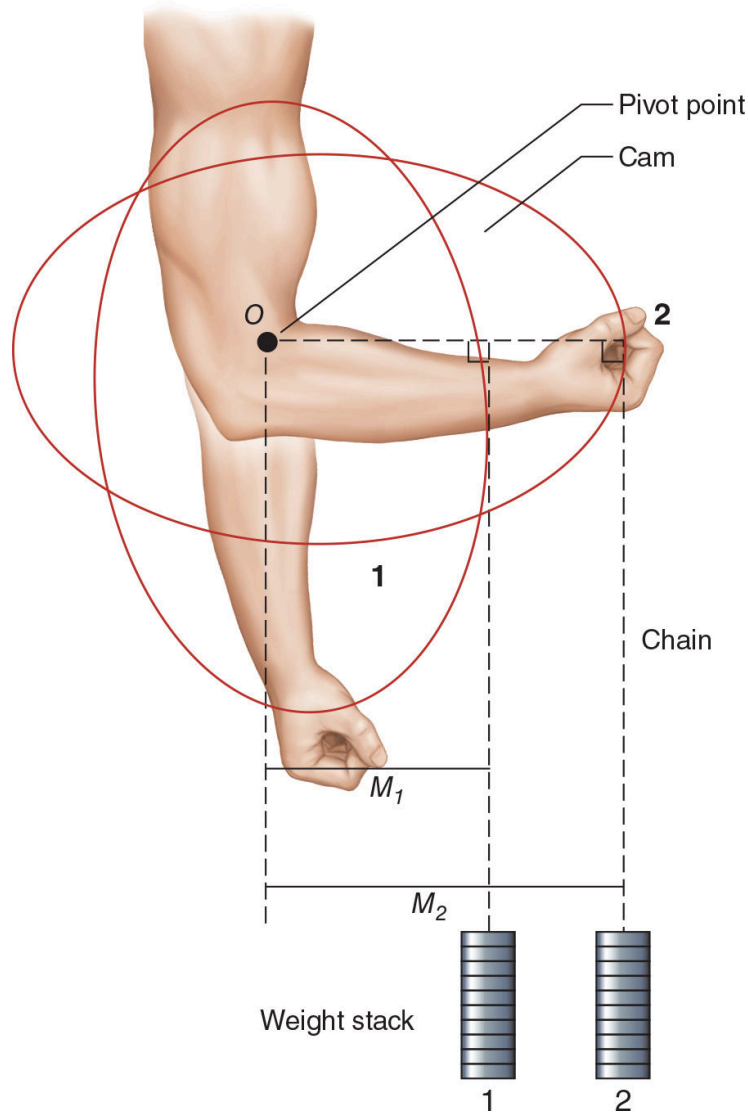


FIGURE 2.14 In cam-based weight-stack machines, the moment arm (M) of the weight stack (length between the chain and the cam axis of rotation) varies during the exercise movement. When the cam is rotated in the direction shown from position 1 to position 2, the moment arm of the weights, and thus the resistive torque, increases.

Inertia

In addition to gravitational force, a barbell or weight stack, when accelerated, exerts **inertial force** on the athlete. Though the force of gravity acts only downward, inertial force can act in any direction. The upward force an athlete exerts equals the weight lifted plus any inertial force, which is the mass times the upward acceleration of the bar. Horizontal bar acceleration occurs if the athlete exerts net force

on the bar directed to the front, back, left, or right. All exercises involve some acceleration at the beginning to bring the bar from a zero to an upward velocity, as well as some deceleration near the top of the exercise to bring the bar's velocity back to zero so that it does not continue its trajectory and fly out of the athlete's hands. With this acceleration pattern, the agonist muscles receive resistance in excess of bar weight early in the range of motion but resistance less than bar weight toward the end of the range of motion (37). The athlete decelerates the bar by either reducing upward force on the bar to less than bar weight to let some or all of the bar's weight decelerate it or pushing down against the bar using the antagonist muscles. In either case, the deceleration has the effect of providing less resistance to the agonist muscles late in the range of motion.

Compared to a slow exercise with minimal acceleration of a given weight, an exercise involving higher acceleration (an "explosive" exercise) provides greater resistance to the muscles involved early in the range of motion and less resistance to the muscles involved toward the end of the range of motion. However, because of the addition of inertia, heavier weights can be handled in accelerative exercises than in slow exercises, providing near-maximal resistance for all muscles involved in the exercise. During a power clean of a heavy weight, for example, the strong leg, hip, and back muscles accelerate the bar vertically to a high enough velocity that, even though the weaker upper body muscles cannot apply the vertical force necessary to accelerate the bar's weight, the bar's momentum (achieved by applying the necessary force over the available time) means that it continues to travel upward, the force of gravity having to work for longer to decelerate it to zero velocity at the highest bar position.

Although acceleration changes the nature of an exercise and makes resistance patterns less predictable, acceleration in resistance training is not necessarily undesirable. Because acceleration is characteristic of natural movements in sport and daily life, resistance training exercises involving acceleration probably produce desirable

neuromuscular training effects. Weightlifting exercises such as the snatch and the clean and jerk are effective for improving the ability to produce high accelerations against heavy resistance (32).

Acceleration and deceleration are characteristic of virtually all natural movements. For example, sprinting requires the athlete's arms and legs to go through repeated cycles of acceleration and deceleration. Throwing a baseball, discus, shot, or javelin all involve sequences of body movements that accelerate the objects to high release speeds. Because acceleration is a particular kind of movement pattern, training with accelerative movements can provide specificity of training. That is why explosive exercises, such as the power clean and high pull, are used in training for many different sports in which the leg and hip muscles provide force for accelerating the body. The **bracketing technique**, in which the athlete performs the sport movement with less than normal and greater than normal resistance, is another form of acceleration training. According to the force–velocity relationship of muscle, a shot-putter who trains with an extra-heavy shot develops greater forces during the accelerative movement than when using the normal shot because the inertia of the heavier implement forces the muscle to contract at relatively low velocity. When a relatively light shot is used, the lower inertia of the shot enables the putter to accelerate the shot more rapidly and to reach a higher speed of release, thereby training the neuromuscular system to operate within desired acceleration and velocity ranges. Although the principle of increasing or decreasing the load during a movement as described has the theoretical basis for increasing acceleration capacity through the aforementioned methods, one should also consider the influence that such changes in loading have during highly specific or technique-oriented activities such as throwing or sprinting. For example, changing implement loading could have some negative consequences on technique since the body needs time to adjust the motor pattern for that particular movement with the new load.

Overcoming Friction: The Impact of Load on Static and Dynamic Friction

Friction is essential for movement control and provides resistance in sled-based training for horizontal pushing and pulling. Researchers measured the horizontal force needed to overcome static (μ_s) and dynamic (μ_d) friction for sled loads of 44.8, 90, and 136.2 kilograms (98.8, 198.4, 300.3 lb) (3). They found mean forces to overcome μ_s were 206, 374, and 514 N, calculating μ_s (horizontal force $\div$ sled load, mass $\times$ g) as 0.47, 0.42, and 0.39. For μ_d , forces were 154, 289, and 415 newtons, calculating μ_d as 0.35, 0.33, and 0.31. Knowing μ_s and μ_d enables us to calculate the force needed to start and maintain sled movement; this is analogous to barbell mass in traditional resistance exercise. These ratios depend on the load and surface conditions.

Friction

Friction is the force that resists sliding and is encountered when one attempts to move an object whose surface is pressed against the surface of another object. Exercise devices that use friction as the main source of resistance include belt- or brake pad-resisted cycle ergometers and wrist curl devices, but an element of friction exists and contributes to exercise load from weight stack machines. For such devices,

$$F_R = k \times F_N \quad (2.9)$$

where F_R is the resistive force; k is the coefficient of friction for the two contact surfaces; and F_N is the normal force, which presses the objects against each other (perpendicular to desired movement direction).

The coefficients of friction for initiating and for maintaining movement are different. All else being equal, it takes more force to initiate movement between two surfaces in contact than to maintain previously initiated movement. Thus, a friction-resisted exercise device requires a relatively high force to initiate movement and a relatively constant force after movement has begun, no matter what the movement speed. Resistance provided by such devices is

sometimes adjusted through a mechanism that alters the normal force that keeps the friction surfaces in contact with each other.

A weighted sled used in training for football or track is an example of a device that is resisted by both friction and inertia. The resistance due to the sled's inertia is directly proportional to both the sled's mass and its acceleration. The resistance due to the friction between the sled's runners and the ground is proportional to both the friction coefficient between surfaces in contact and the net force (which is the normal force in this example) pressing the sled against the ground, which equals the gravitational force minus any upward force exerted by the individual pushing the sled. Mass can be added to the sled to increase the gravitational force. The friction coefficient varies with the surface on which the sled rests (e.g., sand, bare soil, dry grass, wet grass) and the sled's runners. Thus, for outdoor training, such devices do not provide consistently repeatable resistance. They are nevertheless useful in providing horizontal resistance, which cannot be directly provided by weights. It takes more force to get the sled moving than to keep it moving, because the coefficient of static friction is always greater than the coefficient of sliding friction. Once the sled is moving, the coefficient of sliding friction stays relatively constant. Therefore, one should understand that if velocity is kept relatively consistent during sled pulling and pushing, friction resistance will be relatively consistent too. This will be dictated by a combination of the athlete's capacity to apply consistent impulse ($F \times t$) at the foot-ground interface and maintain posture sufficient to effectively transfer this to the sled.

Fluid Resistance

The resistive force encountered by an object moving through a fluid (liquid or gas), or by a fluid moving past or around an object or through an opening, is called **fluid resistance**. Fluid resistance is a significant factor in such sport activities as swimming, rowing, golf, sprinting, discus throwing, and baseball pitching. (Except for swimming and rowing, in which the fluid is water, all these involve

air resistance.) The phenomenon has become important in resistance training with the advent of hydraulic (liquid) and pneumatic (gas) exercise machines and with the increasing popularity of swimming pool exercise routines, particularly among older people and pregnant women. The two sources of fluid resistance are **surface drag**, which results from the friction of a fluid passing along the surface of an object, and **form drag**, which results from the way in which a fluid presses against the front or rear of an object passing through it. Cross-sectional (frontal) area has a major effect on form drag.

Fluid-resisted exercise machines most often use cylinders in which a piston forces fluid through an opening as the exercise movement is performed. The resistive force is greater when the piston is pushed faster, when the opening is smaller, or when the fluid is more viscous. All else being equal, resistance is roughly proportional to the velocity of piston movement:

$$F_R = k \times v \quad (2.10)$$

where F_R is the resistive force; k is a constant that reflects the physical characteristics of the cylinder and piston, the viscosity of the fluid, and the number, size, and shape of the openings; and v is piston velocity relative to the cylinder.

Because fluid cylinders provide resistance that increases with velocity, they allow an initial rapid acceleration that decreases at higher velocities. Movement velocity is thus kept within an intermediate range. Although such machines limit changes in velocity to a certain extent, they are not isokinetic (constant velocity) as is sometimes claimed. Some machines have adjustment knobs that enable the opening size to be changed. A larger opening enables the user to reach a higher movement velocity before the fluid-resistive force curtails the ability to accelerate.

Fluid-resisted machines do not generally provide an eccentric exercise phase, but they may if they incorporate an internal pump. With a free weight, the agonist muscle group acts concentrically

while raising the weight and eccentrically while lowering it. With fluid-resisted machines without eccentric resistance, a muscle group acts concentrically during performance of the primary exercise movement, and the antagonist muscle group acts concentrically during the return to the starting position. In other words, whereas free weights or weight machines involve alternate concentric and eccentric actions of the same muscle with little or no rest in between, fluid-resisted machines generally involve alternate concentric actions of antagonistic muscle groups; each muscle group rests while its antagonist works. The lack of an eccentric muscle action with fluid-resisted machines means that such exercise probably does not provide optimal specificity of training for the many sport movements that involve eccentric muscle actions (e.g., running, jumping, and throwing).

Elasticity

A number of exercise devices, particularly those designed for home use, have elastic components such as springs, bands, bows, or rods as their source of resistance. The resistance provided by a standard elastic component is proportional to the distance it is stretched:

$$F_R = k \times x \quad (2.11)$$

where F_R is the resistive force, k is a constant that reflects the physical characteristics of the elastic component, and x is the length that the elastic component is stretched to beyond its resting length.

The most obvious characteristic of elastic resistance is that the more the elastic component is stretched, the greater the resistance. The problem with devices using elastic resistance is that every exercise movement begins with low resistance and ends with high resistance. This is contrary to the force capability patterns of virtually all human muscle groups, which show a substantial drop-off in force capability toward the end of the range of motion. Another problem with elasticity-resisted machines is that the adjustability of resistance is usually limited by the number of elastic components available to

provide resistance to a movement. An effective resistance exercise device should incorporate enough variation in resistive force that the number of repetitions the athlete can perform is kept within a desirable range.

Certain products provide resistance to vertical jumping with elastic bands as a means of developing jumping power. However, the elastic bands provide little resistance early in the jump when the large gluteal and quadriceps muscles are capable of exerting great force. The bands provide the greatest resistance while the jumper is in the air—serving mainly to pull the jumper back to the ground, rather than resist the muscles, and so increasing the jumper's landing velocity, which may increase injury risk.

Joint Biomechanics: Concerns in Resistance Training

As with any physical activity, resistance training involves a degree of risk. However, the risks involved are generally lower than for many other sport and physical conditioning activities (48, 49). Rates of injury are the highest for team sports; intermediate for running and aerobics; and lowest for cycling, walking, and resistance training, the latter of which has about 4 injuries per 1,000 hours of participation. A study of collegiate American football players showed only 0.35 resistance training–related injuries per 100 players per season. Injuries due to resistance training accounted for only 0.74% of the in-season injury-related time loss of the players (61). Despite the relatively low risk of resistance training, it is desirable to minimize the likelihood of injury through prudent risk management. The following are several factors to consider in avoiding resistance training injuries, with particular attention given to the back, shoulders, and knees.

The risk of injury from resistance training is low compared to that of other sport and physical conditioning activities.

Back

In contrast to quadrupeds, whose vertebral columns hang like the cables on a suspension bridge, humans normally stand upright, with the vertebral bones stacked one on top of another, separated by rubbery disks. The advantage we gain from our upright posture and free use of the arms and hands is accompanied by the disadvantage of having our intervertebral disks under compressive force even when we are merely standing, sitting, walking, or running—and under even more compressive force when we are lifting and carrying (19). When we are in a standing position, any force we exert with

the upper body must be transmitted through the back to the legs and ground. In addition, the back muscles act at a great mechanical disadvantage and must generate forces much greater than the weight of an object lifted. It is for these reasons that the back is particularly vulnerable to injury. It should be noted, however, that spinal internal loads are quite variable with different postures during the movement (31) and that deep squatting positions with load are not necessarily associated with back injury (22).

Back Injury

Back injury can be extremely debilitating, persistent, and difficult to remedy. Thus, every effort should be made to avoid back injury during resistance training. The lower back is particularly vulnerable. It has been observed that 85% to 90% of all intervertebral disk herniations occur at the disk between the lowest two lumbar vertebrae (L4 and L5) or between the lowest lumbar and the top sacral vertebra (L5 and S1) (1, 4). This is not surprising, given the extremely high compressive forces on the disks during lifting. When a weight is supported in the hands or on the shoulders and the trunk is inclined forward, there is great torque about the lower intervertebral disks due to the large perpendicular distance between the lower back and the weight. The back muscles operate at an extremely low mechanical advantage because the perpendicular distance from the line of action of the spinal erector muscles to the intervertebral disks is much shorter (about 5 cm, or 2 in) than the perpendicular distance from the weight to the disks. As a result, the muscles must exert forces that frequently exceed 10 times the weight lifted (4). These forces act to squeeze the intervertebral disks between the adjacent vertebral bodies and can lead to injury.

The neutral-back lifting posture has been found to be better overall than a rounded (opposite of arched) back in minimizing L5/S1 compressive forces and ligament strain (2). Therefore, a normal **lordotic** lumbar spine position is superior to a rounded back for avoiding injury to vertebrae, disks, facet joints, ligaments, and

muscles of the back. In addition, the low back muscles are capable of exerting considerably higher forces when the back is arched rather than rounded (9).

The **vertebral column** is naturally S-shaped, being slightly rounded (**kyphotic**) in the thoracic spine and lordotic in the lumbar spine. The wedge shape of the vertebrae gives the spine its natural curve. However, the intervertebral disks are flat when the back is in its S shape. When the lower back is rounded, the **ventral** (toward the anterior) edges of the vertebral bodies squeeze the front portions of the intervertebral disks. In contrast, extreme arching of the back results in squeezing the **dorsal** (toward the posterior) portions of the disks. Such uneven squeezing of the intervertebral disks likely increases the risk of disk rupture (4). Thus, resistance training exercises should generally be performed with the lower back in a moderately arched position to reduce risk of damage to the disks.

Intra-Abdominal Pressure and Lifting Belts

When the diaphragm and the deep muscles of the torso contract, pressure is generated within the abdominal cavity. Because the abdomen is composed mainly of fluid and normally contains very little gas, it is virtually incompressible (4). The abdominal fluids and tissue kept under pressure by tensing surrounding muscle (deep abdominal muscles and diaphragm) have been described as a **fluid ball** ([figure 2.15](#)) that aids in supporting the vertebral column during resistance training (4). Such support may significantly reduce both the forces required by the erector spinae muscles to perform an exercise and the associated compressive forces on the disks (4, 40).

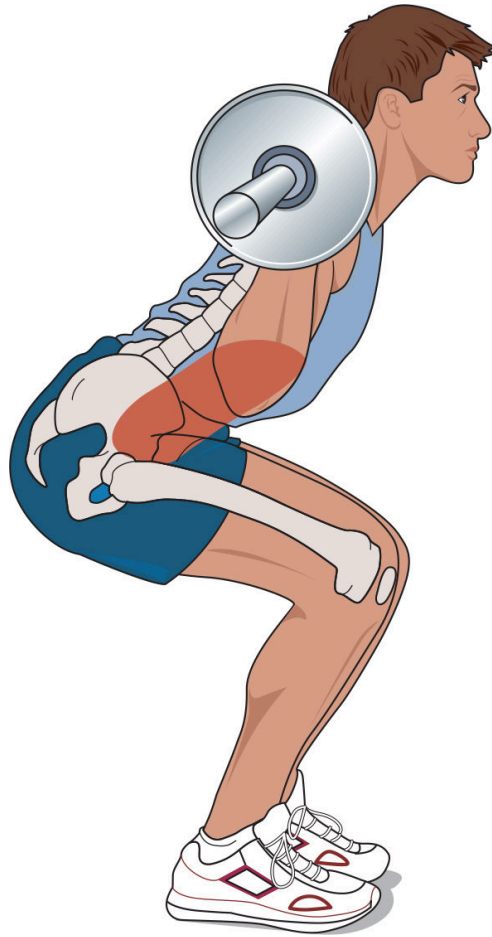


FIGURE 2.15 The fluid ball resulting from contraction of the deep abdominal muscles and the diaphragm.

It is important to note that the **Valsalva maneuver** is not necessary for generation of intra-abdominal pressure. In the Valsalva maneuver, the glottis is closed, thus keeping air from escaping the lungs, and the muscles of the abdomen and rib cage contract, creating rigid compartments of liquid in the lower torso and air in the upper torso. An advantage of the Valsalva maneuver is that it increases the rigidity of the entire torso, making it easier to support heavy loads (20). For example, when lifting heavy loads in the back squat exercise, many athletes use the Valsalva maneuver, particularly when the trunk is most inclined forward, near the transition from the eccentric movement phase to the concentric movement phase. However, pressure in the chest associated with the Valsalva

maneuver can have the undesirable side effect of exerting compressive force on the heart, making it more difficult for blood to return to the heart. Also, the Valsalva maneuver can transiently raise blood pressure to slightly elevated levels (20). The diaphragm and the abdominal muscles can contract *without* the glottis being closed, however, creating the fluid ball in the abdomen without pressurizing the chest compartment. This must be regarded as the safer way, of the two options, to add support to the lower spine without building up pressure in the chest, and it is the technique that should be used for most resistance training. One can build up intra-abdominal pressure without building up chest pressure by consciously keeping the airway open. During a strenuous repetition, the abdominal muscles and diaphragm contract reflexively, even with the airway open. Athletes, particularly those who compete in weightlifting or powerlifting, may choose to use the Valsalva maneuver if they recognize and accept the risks involved and have the experience to avoid increasing pressure to the point of blackout.

Weightlifting belts have been shown to increase intra-abdominal pressure during resistance training and are therefore probably effective in improving safety when used correctly (21, 38, 62). It has been cautioned, however, that if an athlete performs all of the exercises with a belt, the abdominal muscles that produce intra-abdominal pressure might not get enough training stimulus to develop optimally (21). It is particularly risky for an individual who has become accustomed to wearing a belt to suddenly perform an exercise without one, because the abdominal musculature might not be capable of generating enough intra-abdominal pressure to significantly reduce erector spinae muscle forces. The resulting excessive compressive forces on the disks could increase the chance of back injury. Conservative recommendations are as follows:

- A weight belt is not needed for exercises that do not directly affect the lower back.
- For exercises directly stressing the back, an individual should refrain from wearing a belt during lighter sets but may wear

one for near-maximal and maximal sets. The beltless sets allow the deep abdominal muscles, which generate intra-abdominal pressure, to receive a training stimulus without placing excessive compressive forces on the intervertebral disks.

- Individuals may reasonably choose never to wear lifting belts if they build up the strength of their back muscles and the muscles that generate intra-abdominal pressure in a gradual and systematic manner and if they practice safe resistance training exercise techniques. Many world-class weightlifters never wear belts.

Shoulders

The shoulder is particularly prone to injury during resistance training, due to both its structure and the forces to which it is subjected during a training session. Like the hip, the shoulder is capable of rotating in any direction. The hip is a stable ball-and-socket joint, but the glenoid cavity of the shoulder, which holds the head of the humerus, is not a true socket and is significantly less stable. The shoulder joint has the greatest range of motion of all the joints in the human body, but the joint's excessive mobility contributes to its vulnerability, as does the proximity of the bones, muscles, tendons, ligaments, and bursae in the shoulder.

The stability of the shoulder largely depends on the glenoid labrum; the joint synovium; and capsules, ligaments, muscles, tendons, and bursae. The rotator cuff muscles (supraspinatus, infraspinatus, subscapularis, and teres minor) and the pectorals are particularly instrumental in keeping the ball of the humerus in place. With the shoulder's great range of motion, its various structures can easily impinge on one another, causing tendinitis as well as inflammation and degeneration of contiguous tissue. High forces generated during resistance training can result in tearing of ligaments, muscles, and tendons. Athletes must take particular care when performing the various forms of the bench and shoulder press

exercises because of the great stresses placed on the shoulder. For these exercises, it is particularly important to warm up with relatively light weights and to follow a program that exercises the shoulder in a balanced way, using all of its major movements.

Knees

The knee is prone to injury because of its location between two long levers (the upper and lower leg). Flexion and extension about the knee occur almost exclusively in the sagittal plane. Rotation in the frontal plane and transverse plane is prevented mainly by ligamentous and cartilaginous stabilizing structures. Frontal plane torque on the knee occurs, for example, when a football player is hit at midleg from the side while the foot is planted firmly on the ground. Fortunately, in training, resistive torques occur almost exclusively within the knee's normal plane of rotation.

Of the various components of the knee, the patella and surrounding tissue are most susceptible to the kinds of forces encountered in resistance training. The patella's main function is to hold the quadriceps tendon away from the knee axis of rotation, thereby increasing the moment arm of the quadriceps group and its mechanical advantage ([figure 2.5, page 30](#)). If inappropriate load, volume, or recovery is introduced, repetitive high forces encountered by the patellar tendon during resistance training (as with any high-force activity such as running) can lead to tendinitis, which is characterized by tenderness and swelling. There is no inherent risk of tendinitis with performing these exercises; rather, tendinitis is simply a function of too much volume and intensity without appropriate progression.

It is not unusual for individuals to use knee wraps during training or competition in order to assist with maximizing performance or preventing injury. Wraps vary from the thin, elastic, pull-on variety that can be purchased in drug stores to the heavy, specialized wraps sold only through weightlifting supply houses. The use of knee wraps, particularly the heavy ones, is most prevalent among

powerlifters. Very little research has been done on the efficacy of knee wraps. Detrimental side effects have been reported, however, including skin damage and chondromalacia patellae, the wearing down and roughening of the posterior surface of the patella (36). Through a spring effect alone, heavy wraps around the knees added an average of 25 pounds (110 N) to squat lifting force. The notion that wraps work only by stabilizing the knee, lessening the athlete's fear of injury, or providing a kinesthetic cue is incorrect (36). The wraps actually provide direct help in extending the knee. On the basis of the lack of evidence that knee wraps prevent injury, athletes should minimize their use. If used at all, knee wraps should be limited to the sets with the heaviest loads.

Elbows and Wrists

The primary concerns with elbow and wrist injury involve overhead exercises (10, 57). However, the risk with overhead lifting is quite small in comparison to the common source of injury of these joints, which includes participation in overhead sports such as throwing events or the tennis serve (10, 57). Other examples of possible injury are elbow dislocation, sometimes observed in gymnastics (39), and overuse-related injuries such as traction apophysitis, sometimes observed in diving, wrestling, and hockey (39). One of the primary concerns is epiphyseal growth plate damage or overuse either in the posterior aspect of the elbow or in the distal radius in young athletes (39). The prevalence of elbow or wrist injury with weightlifting is very sporadic and often referred to in the literature only through case studies. One study indicated a triceps tendon tear in a middle-aged competitive weightlifter (47) and in another study, a bilateral distal biceps tendon rupture in a recreational weight trainer (50). A study examining 245 competitive powerlifters found an extremely low incidence of elbow and wrist injury (52). Only very limited data have been presented to suggest possible distal radial epiphysis fracture in adolescent weightlifters (11, 29). According to a study that surveyed 500 experts in the field of sports medicine, most

respondents indicated that avoiding resistance training before physeal closure was not necessary (43).

Conclusion

It is hoped that readers will apply the biomechanical principles discussed in this chapter to the selection of resistance exercise equipment and the design of exercise programs. Knowledge of how different types of exercise provide specific patterns of resistance to the body can aid in developing safe and effective programs to suit the specific needs of athletes engaged in various sports and others who engage in resistance training for enhancement of physical performance, health, sense of well-being, and self-confidence.

KEY TERMS

acceleration
agonist
anatomical position
angle of pennation
angular displacement
angular velocity
antagonist
biomechanics
bracketing technique
cartilaginous joint
classic formula
concentric muscle action
displacement
distal
dorsal
eccentric muscle action
elasticity
fibrous attachments
first-class lever
fleshy attachments
fluid ball

fluid resistance
force
form drag
friction
frontal plane
fulcrum
gravity
impulse
inertia
inertial force
insertion
isometric muscle action
kinetic energy
kyphotic
lever
lordotic
mass
mechanical advantage
mechanical energy
moment
moment arm
momentum
muscle force
origin
peak power
pennate muscle
potential energy
power
proximal
rate coding
recruitment
resistive force
sagittal plane
scalar

second-class lever
sticking point
strength
structural exercise
surface drag
synergist
tendons
third-class lever
titin
torque
transverse plane
Valsalva maneuver
vector
velocity
ventral
vertebral column
work

STUDY QUESTIONS

1. Gravity on the surface of the moon is approximately $\frac{1}{6}$ that on the surface of the earth. Considering a kettlebell, labeled 30 kilograms (66 lb), what would using the kettlebell feel like (approximately) on the surface of the moon?
 - a. 5 kg (11 lb)
 - b. 15 kg (33 lb)
 - c. 30 kg (66 lb)
 - d. 180 kg (397 lb)
2. In sliding-filament theory, what is the third contractile element used in muscle contraction?
 - a. myosin
 - b. actin
 - c. titin
 - d. sarcomere

3. What class of lever is the lead forearm in a baseball pitch at ball release?
 - a. first-class lever
 - b. second-class lever
 - c. third-class lever
 - d. fourth-class lever
4. For a system-wide view of an athlete, which of the following aspects of fitness and performance is the most relevant?
 - a. impulse
 - b. peak force
 - c. mean force
 - d. rate of force development
5. A sidestepping (cutting) movement involves ankle, knee, and hip movement primarily in which of the following anatomical planes?
 - a. sagittal
 - b. perpendicular
 - c. frontal
 - d. transverse